

Container-based High-Performance Computing Approaches for Simulating Large-Scale Biological Neural Networks

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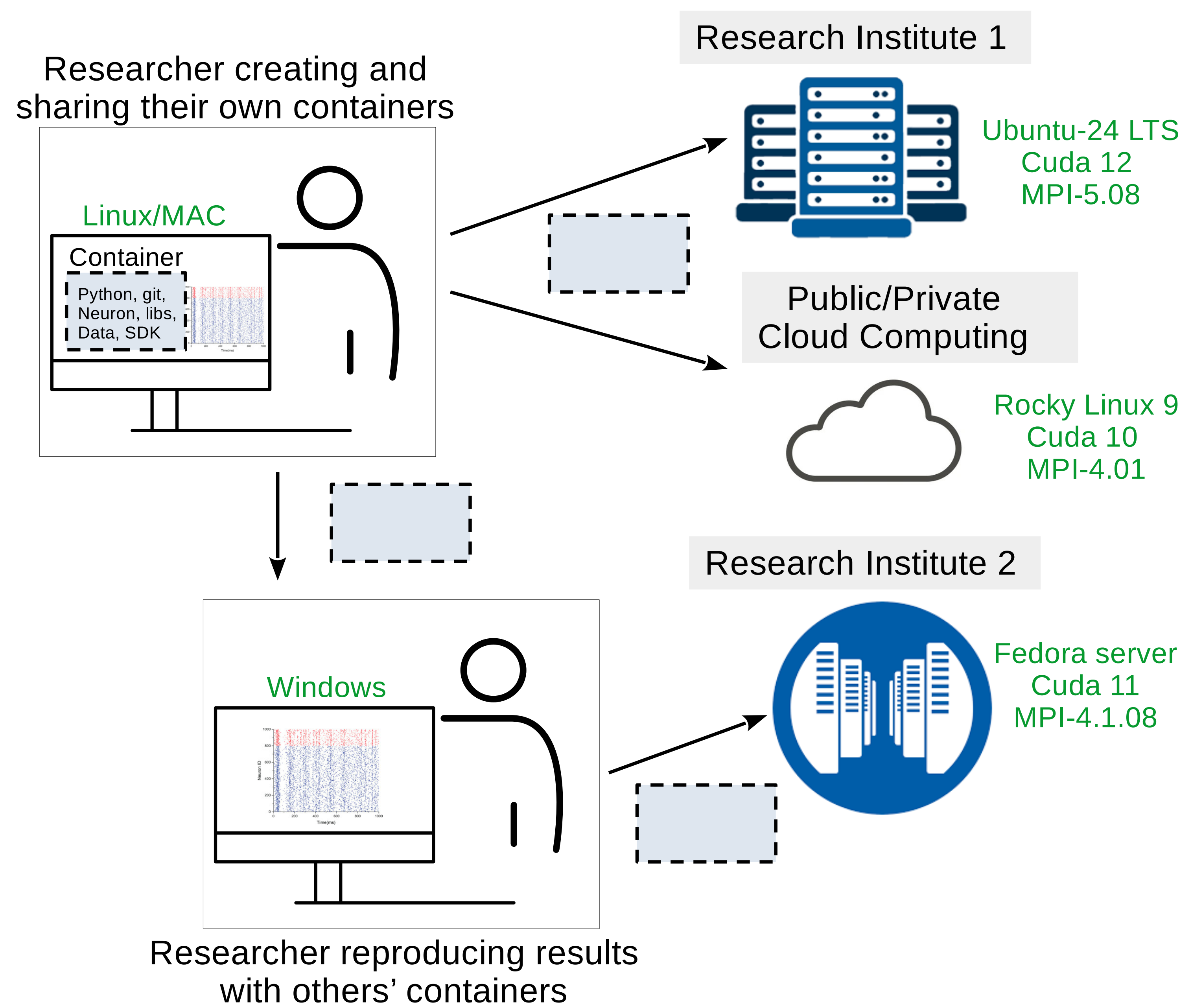
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Introduction

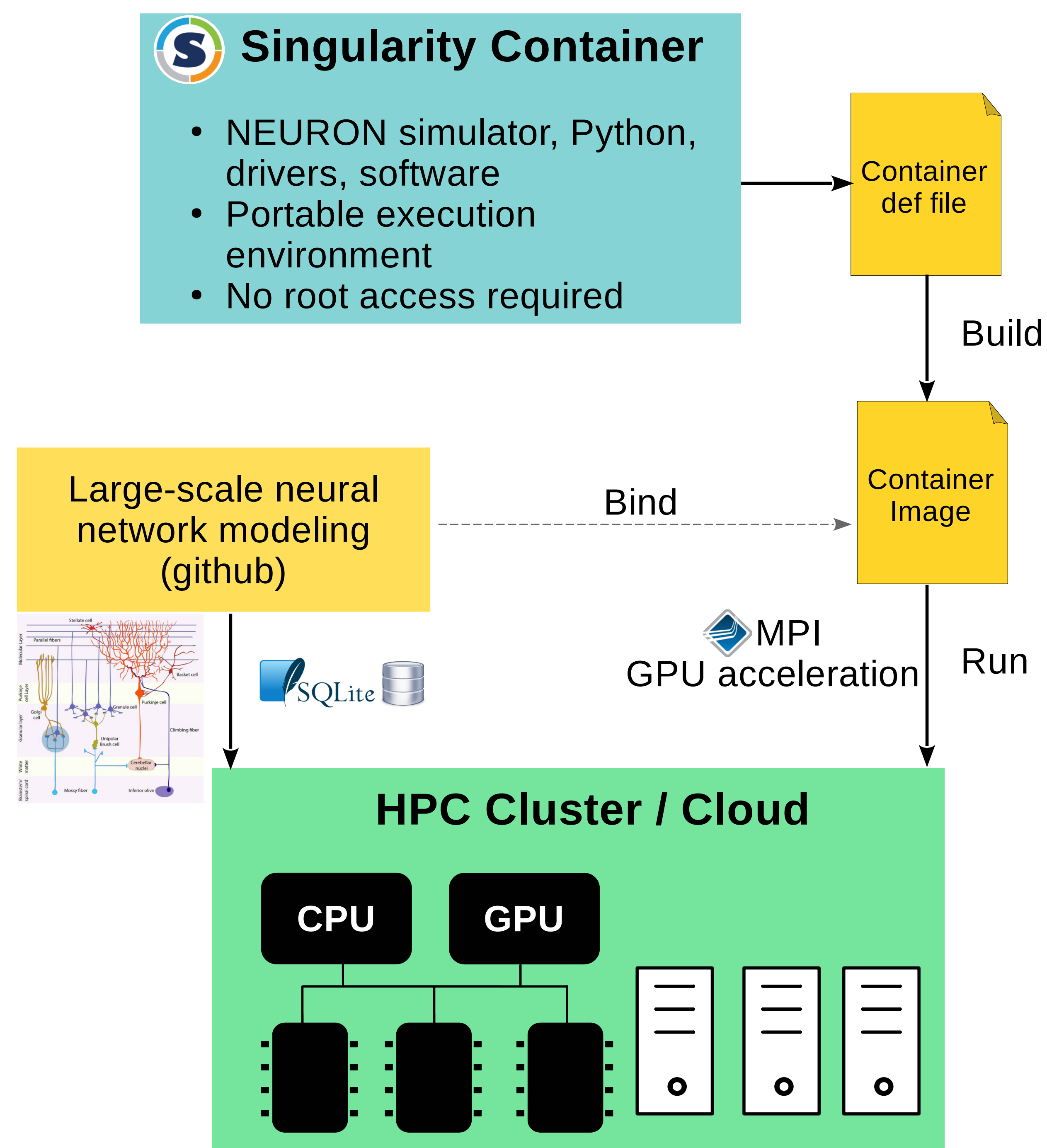
1. Modeling large-scale, biophysically detailed neural networks increasingly relies on high-performance computing (HPC).
2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to software incompatibilities and complex dependencies.
3. We address this issue using singularity containers that encapsulate the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on HPC/Cloud systems.
4. We give a practical demonstration by benchmarking HPC environments with cerebellar mossy fiber–granule cell network simulations.

Containerization for Scientific Computing

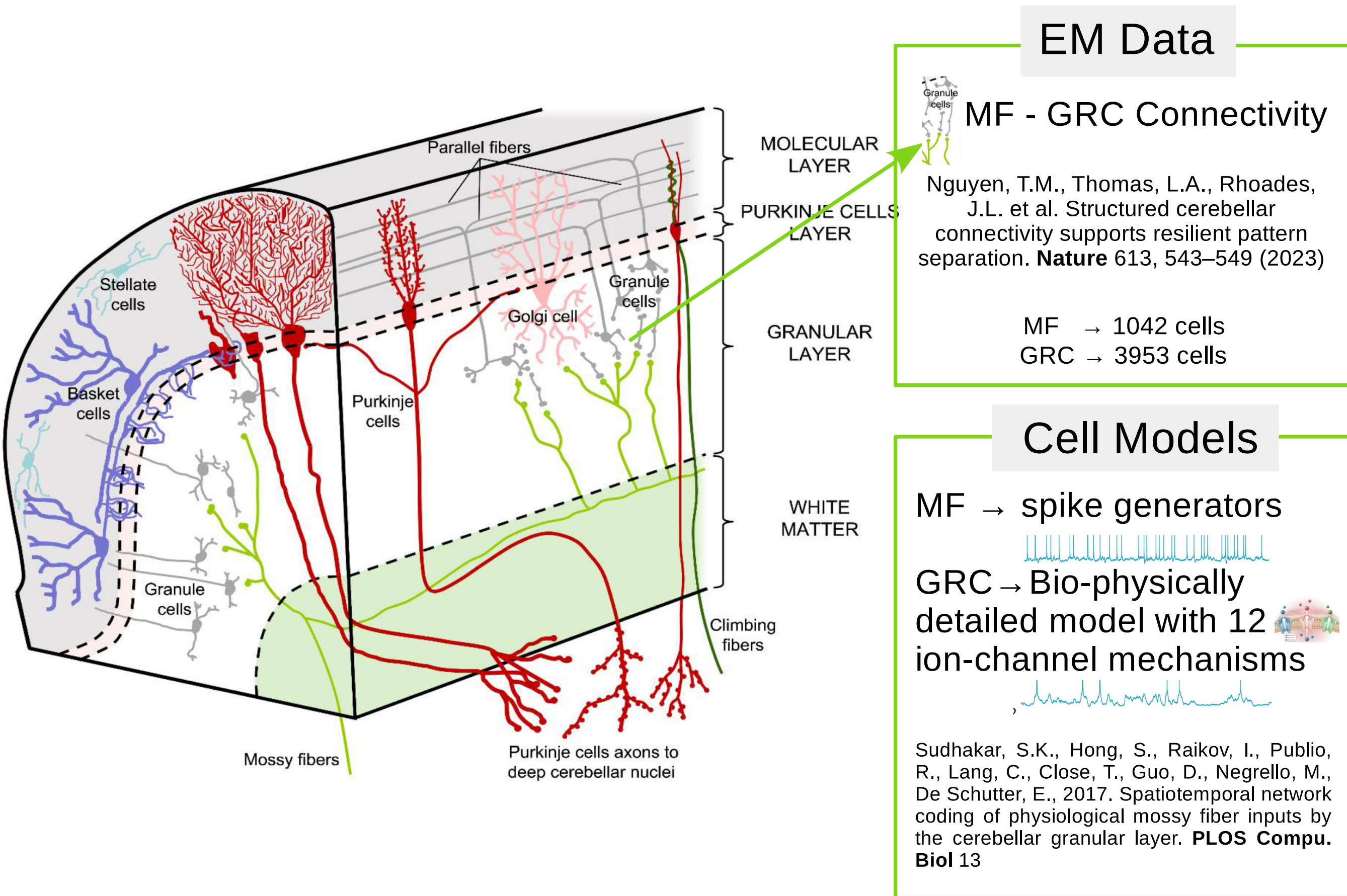
Container is a lightweight, standalone, and executable package. It is designed to include everything needed to run a scientific software.



Our Computational Framework



Benchmarking Data and Network: Cerebellar Mossy Fiber → Granule Cell



We run multiple ($n = 80$) copies of the network connectivity model simultaneously to simulate extreme computational workloads. This allows us to rigorously assess the scalability and performance limits of GPUs under high-demand conditions.

Table: Network Configuration and Simulation Parameters

Number of MF	85,600
Number of GRC	314,000
Number of compartments	942,000
Number of synapses	990,960
Simulation time	2 s
Simulation software	CORENeuron

Benchmarking HPC Environments

Table: Performance in CPU parallel computing (30-task MPI)

CPU	Core	Solver Time (s)	Simulator
Intel(R)-Xeon	30	564	CORENeuron-CPU

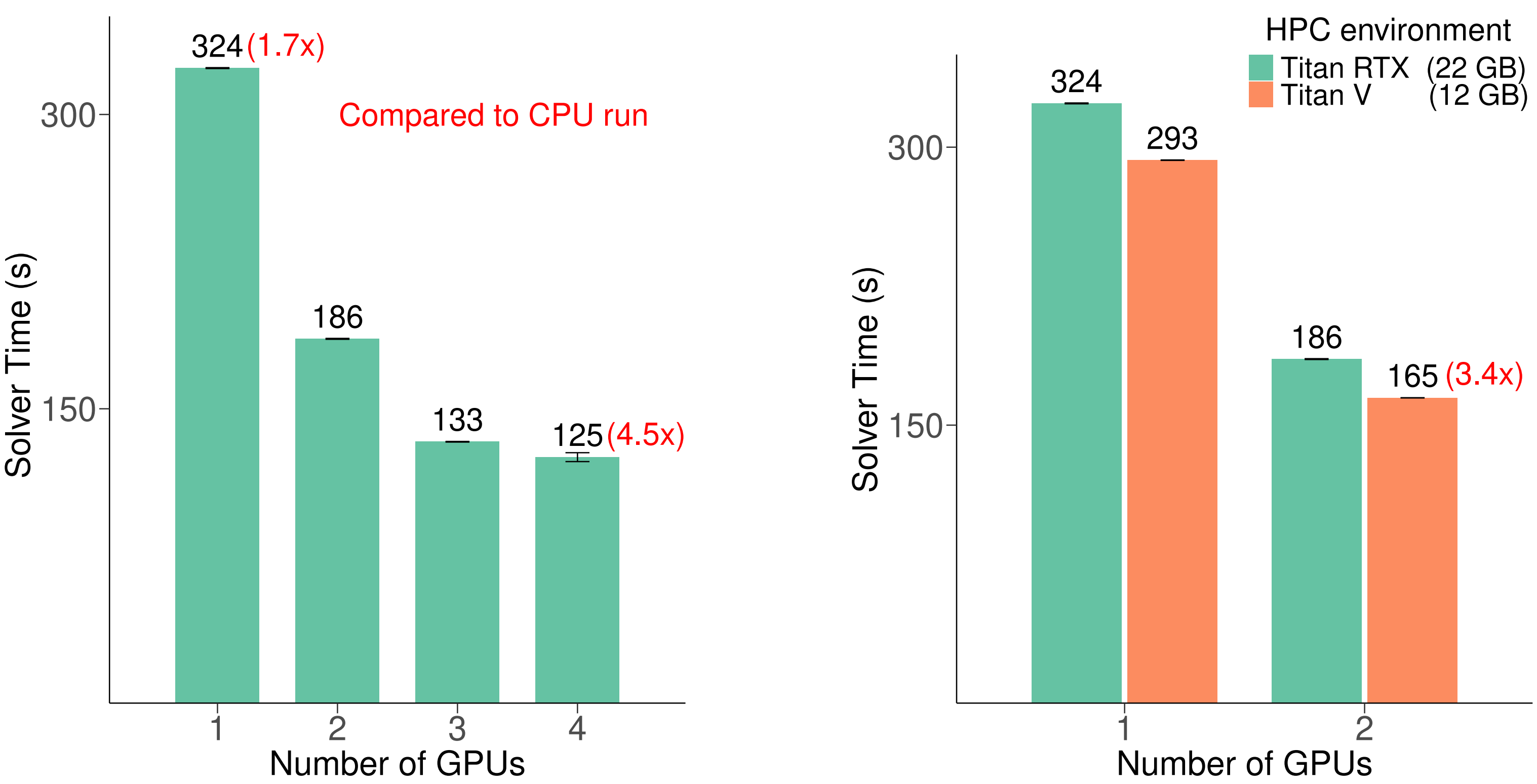


Figure: Solver time comparisons in CORENeuron simulations across HPC platforms.

TITAN RTX outperform 30-core CPUs by up to 4.5x, with performance improving as GPU count increases.

Future works:

- 🔧 Develop a biologically realistic cerebellar circuit model including granule layer (with Golgi cells), Purkinje layer, and associated interneurons.
- 🔧 Benchmark network performance across multiple institutions to evaluate and understand the scalability and performance limits of HPC systems.

Acknowledgment

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